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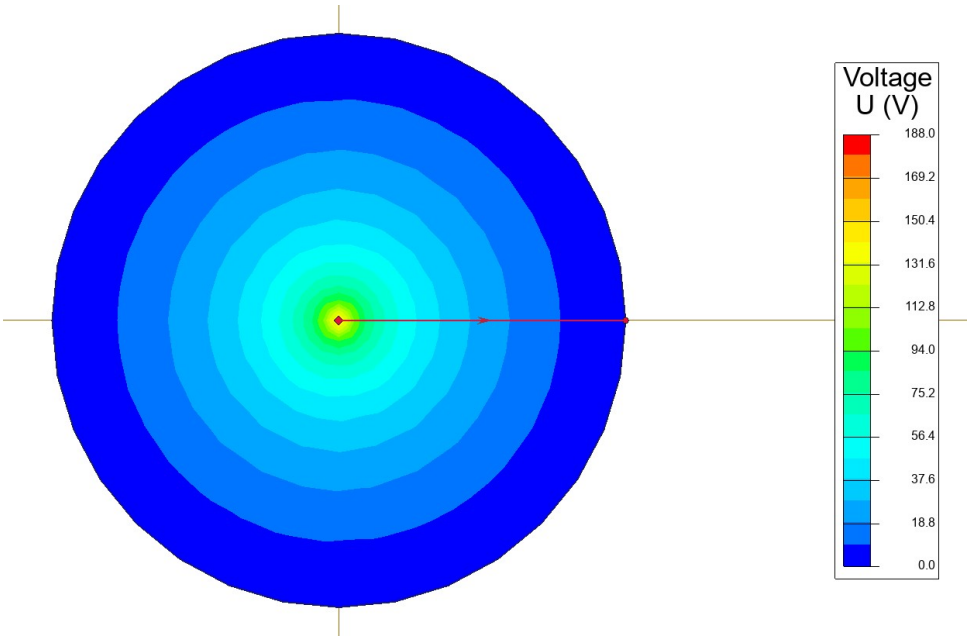
5. Revisión. **11**

1. Hilo Infinito.

Hacemos el **Hilo Infinito** con el mallado y la solución del mismo.

Configuración en **QuickField**:

- $\lambda = 2 \times 10^{-9}$.
- $\epsilon_r = 1$.
- V = 0.



Solucionamos y obtenemos los resultados.

L, mm	x, mm	y, mm	N_x	N_y	U, V	E, V/m	E_x, V/m	E_y, V/m	E_n, V/m	E_t, V/m	D, C/m^2	D_x, C/m^2	D_y, C/m^2	D_n, C/m^2	D_t, C/m^2	ε,	w, J/m^3
0	0	0	0	-1	188.287	0.0271454	-0.0269647	0.00312618	-0.00312618	-0.0269647	2.4035e-13	-2.38751e-13	2.76798e-14	-2.76798e-14	-2.38751e-13	1	3.2622e-15
0.05	0.05	0	0	-1	107.808	792677	792677	-915.341	915.341	792677	7.01851e-6	7.01851e-6	-8.1046e-9	8.1046e-9	7.01851e-6	1	2.78171
0.1	0.1	0	0	-1	82.6697	370061	370061	92.0018	-92.0018	370061	3.27659e-6	3.27659e-6	8.14601e-10	-8.14601e-10	3.27659e-6	1	0.606268
0.15	0.15	0	0	-1	68.2464	253981	253981	-13.7297	13.7297	253981	2.24879e-6	2.24879e-6	-1.21565e-10	1.21565e-10	2.24879e-6	1	0.285575
0.2	0.2	0	0	-1	57.9303	186573	186573	9.76938	-9.76938	186573	1.65195e-6	1.65195e-6	8.64999e-11	-8.64999e-11	1.65195e-6	1	0.154104
0.25	0.25	0	0	-1	49.7499	145316	145316	17.7015	-17.7015	145316	1.28666e-6	1.28666e-6	1.56733e-10	-1.56733e-10	1.28666e-6	1	0.0934858
0.3	0.3	0	0	-1	43.4442	122942	122942	0.0658306	-0.0658306	122942	1.08855e-6	1.08855e-6	5.82876e-13	-5.82876e-13	1.08855e-6	1	0.0669147
0.35	0.35	0	0	-1	37.8631	105335	105335	-14.5915	14.5915	105335	9.32655e-7	9.32655e-7	-1.29196e-10	1.29196e-10	9.32655e-7	1	0.0491206
0.4	0.4	0	0	-1	32.9676	92239.1	92239.1	-26.4294	26.4294	92239.1	8.16703e-7	8.16703e-7	-2.34011e-10	2.34011e-10	8.16703e-7	1	0.037666
0.45	0.45	0	0	-1	28.8175	82162.7	82162.7	27.0425	-27.0425	82162.7	7.27484e-7	7.27484e-7	2.3944e-10	-2.3944e-10	7.27484e-7	1	0.029886
0.5	0.5	0	0	-1	24.9796	73351.7	73351.6	107.886	-107.886	73351.6	6.49469e-7	6.49469e-7	9.5524e-10	-9.5524e-10	6.49469e-7	1	0.0238198
0.55	0.55	0	0	-1	21.5768	66211.9	66211.9	104.233	-104.233	66211.9	5.86253e-7	5.86252e-7	9.22902e-10	-9.22902e-10	5.86252e-7	1	0.0194085
0.6	0.6	0	0	-1	18.5593	60552.5	60552.5	25.743	-25.743	60552.5	5.36143e-7	5.36143e-7	2.27933e-10	-2.27933e-10	5.36143e-7	1	0.0162324
0.65	0.65	0	0	-1	15.5417	54893.1	54893.1	-52.7474	52.7474	54893.1	4.86034e-7	4.86034e-7	-4.67036e-10	4.67036e-10	4.86034e-7	1	0.01334
0.7	0.7	0	0	-1	13.0154	51735.9	51735.9	-21.7711	21.7711	51735.9	4.58079e-7	4.58079e-7	-1.92765e-10	1.92765e-10	4.58079e-7	1	0.0118496
0.75	0.75	0	0	-1	10.5336	48805.6	48805.6	19.1348	-19.1348	48805.6	4.32134e-7	4.32134e-7	1.69423e-10	-1.69423e-10	4.32134e-7	1	0.0105453
0.8	0.8	0	0	-1	8.05175	45875.4	45875.4	60.0407	-60.0407	45875.4	4.0619e-7	4.06189e-7	5.31611e-10	-5.31611e-10	4.06189e-7	1	0.00931707
0.85	0.85	0	0	-1	5.95521	43881.6	43881.6	55.0007	-55.0007	43881.6	3.88536e-7	3.88536e-7	4.86987e-10	-4.86987e-10	3.88536e-7	1	0.00852481
0.9	0.9	0	0	-1	3.97014	42158.8	42158.8	36.6672	-36.6672	42158.8	3.73282e-7	3.73282e-7	3.24658e-10	-3.24658e-10	3.73282e-7	1	0.00786856
0.95	0.95	0	0	-1	1.98507	40435.9	40435.9	18.3336	-18.3336	40435.9	3.58027e-7	3.58027e-7	1.62329e-10	-1.62329e-10	3.58027e-7	1	0.00723859
1	1	0	0	-1	0	38713.1	38713.1	0	0	38713.1	3.42773e-7	3.42773e-7	0	0	3.42773e-7	1	0.0066349

Modificamos el script en **Python** para nuestro ejemplo.

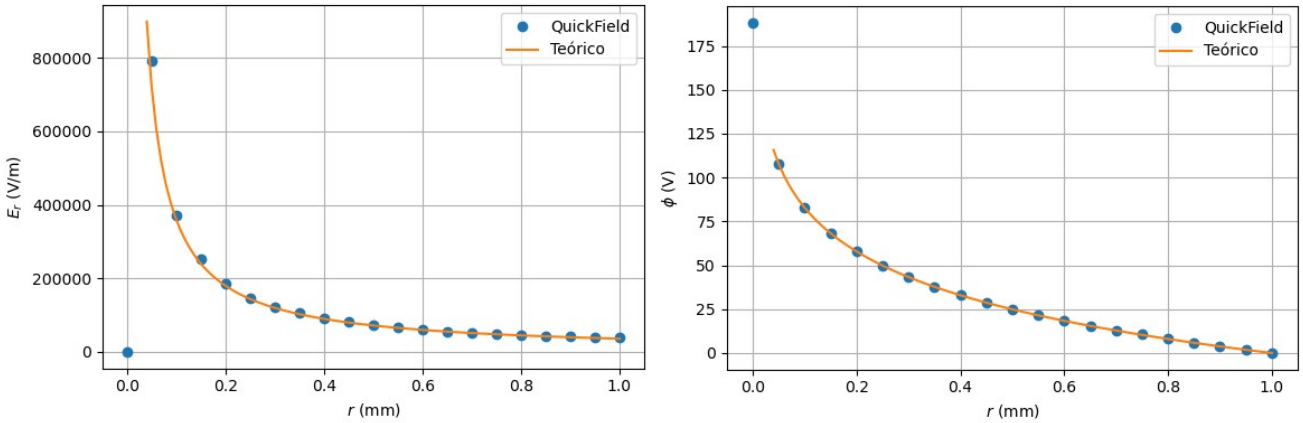
```
# Extraer columnas de interés
L = data["L_mm"]
E = data["E_V/m"]
U = data["U_V"]

##############################################################################################
# Resultado teórico: campo y potencial de un plano cargado infinito
eps0 = 8.854e-12  # permitividad del vacío
lambda_carga = 2e-9 # C/m
r = np.linspace(0.04,np.max(L),101) # malla de puntos del resultado teórico


E_teo = lambda_carga / (2 * np.pi * eps0 * r * 1e-3)
r_0 = 1  # posición del origen de potencial (mm)

U_teo = lambda_carga / (2 * np.pi * eps0) * np.log(r_0/r)
```

Obtenemos el plot de las figuras.



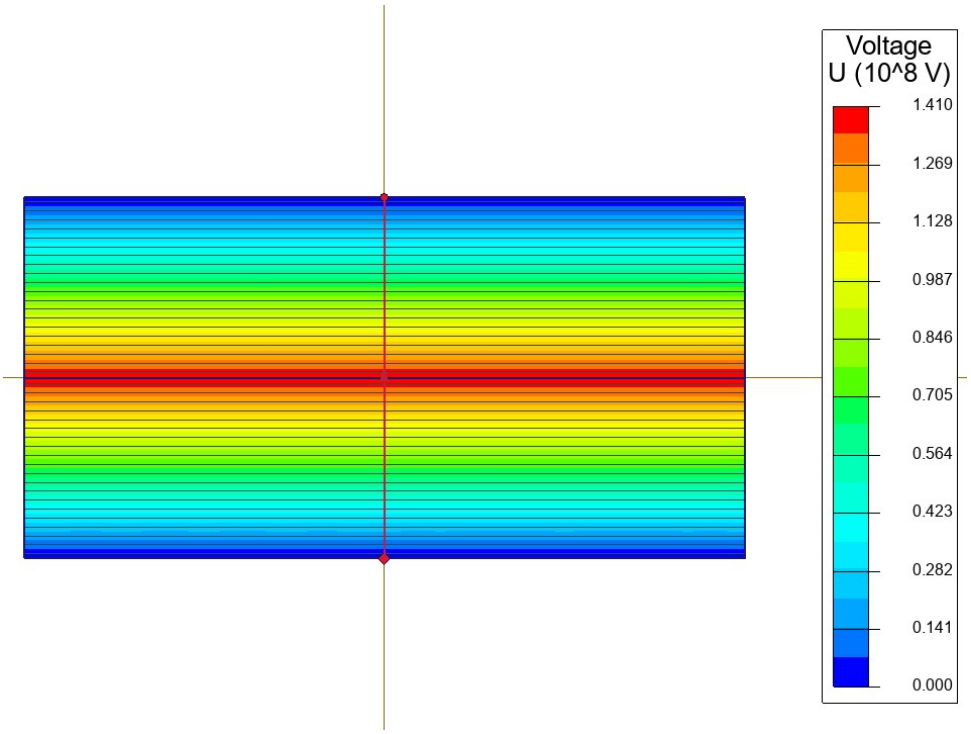
1.1. Conclusión.

Lo resultados en **QuickField** validan la dependencia radial teórica.

2. Plano Cargado.

Configuración en **QuickField**:

- Vacío $\epsilon_r = 1$.
- Contorno V $= 0$.
- Carga $\sigma = 0.5\, C/m^2$.



Solucionamos el plano.

L, mm	x, mm	y, mm	N_x	N_y	U, V	E, V/m	E_x, V/m	E_y, V/m	E_n, V/m	E_t, V/m	D, C/m^2	D_x, C/m^2	D_y, C/m^2	D_n, C/m^2	D_t, C/m^2	ε_r	w, J/m^3
0	0	-5	1	-0	0	2.82352e+10	0	-2.82352e+10	0	-2.82352e+10	0.25	0	-0.25	0	-0.25	1	3.5294e+9
0.5	-1.51788e-15	-4.5	1	-0	14117600	2.82352e+10	37.0832	-2.82352e+10	37.0832	-2.82352e+10	0.25	3.28342e-10	-0.25	3.28342e-10	-0.25	1	3.5294e+9
1	-3.03577e-15	-4	1	-0	28235200	2.82352e+10	74.1665	-2.82352e+10	74.1665	-2.82352e+10	0.25	6.56684e-10	-0.25	6.56684e-10	-0.25	1	3.5294e+9
1.5	-3.03577e-15	-3.5	1	-0	42352800	2.82352e+10	196.254	-2.82352e+10	196.254	-2.82352e+10	0.25	1.73767e-9	-0.25	1.73767e-9	-0.25	1	3.5294e+9
2	-3.03577e-15	-3	1	-0	56470400	2.82352e+10	318.342	-2.82352e+10	318.342	-2.82352e+10	0.25	2.81866e-9	-0.25	2.81866e-9	-0.25	1	3.5294e+9
2.5	-3.03577e-15	-2.5	1	-0	70588100	2.82352e+10	267.96	-2.82352e+10	267.96	-2.82352e+10	0.25	2.37257e-9	-0.25	2.37257e-9	-0.25	1	3.5294e+9
3	-3.03577e-15	-2	1	-0	84705700	2.82352e+10	217.578	-2.82352e+10	217.578	-2.82352e+10	0.25	1.92648e-9	-0.25	1.92648e-9	-0.25	1	3.5294e+9
3.5	-3.03577e-15	-1.5	1	-0	98823300	2.82352e+10	61.3016	-2.82352e+10	61.3016	-2.82352e+10	0.25	5.42776e-10	-0.25	5.42776e-10	-0.25	1	3.5294e+9
4	-3.03577e-15	-1	1	-0	1.12941e+8	2.82352e+10	-94.9748	-2.82352e+10	-94.9748	-2.82352e+10	0.25	-8.40925e-10	-0.25	-8.40925e-10	-0.25	1	3.5294e+9
4.5	-3.03577e-15	-0.5	1	-0	1.27058e+8	2.82352e+10	-137.535	-2.82352e+10	-137.535	-2.82352e+10	0.25	-1.21776e-9	-0.25	-1.21776e-9	-0.25	1	3.5294e+9
5	-3.03577e-15	0	1	-0	1.41176e+8	2.82352e+10	-180.095	-2.82352e+10	-180.095	-2.82352e+10	0.25	-1.59459e-9	-0.25	-1.59459e-9	-0.25	1	3.5294e+9
5.5	-2.42861e-15	0.5	1	-0	1.27058e+8	2.82352e+10	-269.538	2.82352e+10	-269.538	2.82352e+10	0.25	-2.38654e-9	0.25	-2.38654e-9	0.25	1	3.5294e+9
6	-1.82146e-15	1	1	-0	1.12941e+8	2.82352e+10	-329.21	2.82352e+10	-329.21	2.82352e+10	0.25	-2.91488e-9	0.25	-2.91488e-9	0.25	1	3.5294e+9
6.5	-1.21431e-15	1.5	1	-0	98823300	2.82352e+10	-404.829	2.82352e+10	-404.829	2.82352e+10	0.25	-3.58444e-9	0.25	-3.58444e-9	0.25	1	3.5294e+9
7	-6.07153e-16	2	1	-0	84705700	2.82352e+10	-480.449	2.82352e+10	-480.449	2.82352e+10	0.25	-4.25399e-9	0.25	-4.25399e-9	0.25	1	3.5294e+9
7.5	3.08149e-30	2.5	1	-0	70588100	2.82352e+10	-203.693	2.82352e+10	-203.693	2.82352e+10	0.25	-1.80354e-9	0.25	-1.80354e-9	0.25	1	3.5294e+9
8	6.07153e-16	3	1	-0	56470400	2.82352e+10	73.0631	2.82352e+10	73.0631	2.82352e+10	0.25	6.46914e-10	0.25	6.46914e-10	0.25	1	3.5294e+9
8.5	1.21431e-15	3.5	1	-0	42352800	2.82352e+10	14.5664	2.82352e+10	14.5664	2.82352e+10	0.25	1.28974e-10	0.25	1.28974e-10	0.25	1	3.5294e+9
9	1.82146e-15	4	1	-0	28235200	2.82352e+10	-43.9302	2.82352e+10	-43.9302	2.82352e+10	0.25	-3.88966e-10	0.25	-3.88966e-10	0.25	1	3.5294e+9
9.5	9.1073e-16	4.5	1	-0	14117600	2.82352e+10	-21.9651	2.82352e+10	-21.9651	2.82352e+10	0.25	-1.94483e-10	0.25	-1.94483e-10	0.25	1	3.5294e+9
10	0	5	1	-0	0	2.82352e+10	0	2.82352e+10	0	2.82352e+10	0.25	0	0.25	0	0.25	1	3.5294e+9

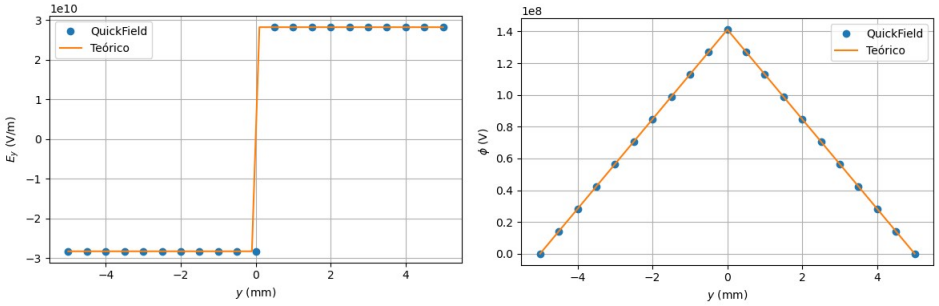
Corregimos σ en el script de Python.

```
###############################################################################################
# Extraer columnas de interés
L = data["y_mm"]
E = data["Ey_V/m"]
U = data["U_V"]

###############################################################################################
# Resultado teórico: campo y potencial de un plano cargado infinito

y = np.linspace(np.min(L),np.max(L),101) # malla de puntos del resultado teórico
eps0 = 8.854e-12  # permitividad del vacío
sigma_carga = 0.5  # C/m^2
E_teo = sigma_carga / (2 * eps0) * np.sign(y)
y_0 = -5 # posición del origen de potencial (mm)
U_teo = -sigma_carga / (2 * eps0) * (np.abs(y) - np.abs(y_0))*1e-3
```

La graficas que obtenemos son:



2.1. Conclusión.

Los resultados en QuickField validan que **el campo eléctrico de un plano infinito es constante e independiente de la distancia**, cumpliendo $E = \sigma/(2\epsilon_0)$.

3. Condensador Plano.

Configuración en **QuickField**:

- **Dimensiones Condensador:**
- $W = 20mm.$
- $h = 20\,mm.$
- $L_z = 10\,mm.$
- Material IZQ. $\epsilon_{r1} = 4.$
- Material DER. $\epsilon_{r2} = 1.$
- Voltaje $V = 10\,V.$
- Separación $d = 20\,mm.$

Condensador Plano y Q_s

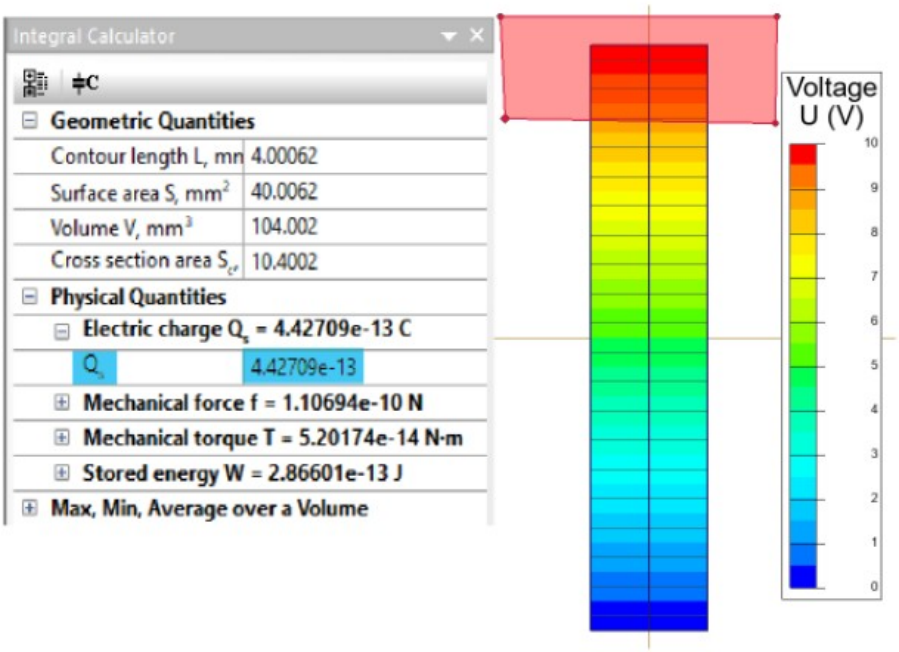


Tabla QuickField.

L, mm	x, mm	y, mm	N_x	N_y	U, V	E, V/m	E_x, V/m	E_y, V/m	E_n, V/m	E_t, V/m	D, C/m²	D_x, C/m²	D_y, C/m²	D_n, C/m²	D_t, C/m²	ε_r	w, J/m³
3.87651	-1.05651	7.41849	-0.0175412	-0.999846	8.70924	500	2.09472e-6	-500	499.923	8.77058	1.77084e-8	7.41881e-17	-1.77084e-8	1.77057e-8	3.10626e-10	4	4.42709e-6
5.16869	0.235458	7.39582	-0.0175412	-0.999846	8.69791	500	-9.62221e-7	-500	499.923	8.77058	4.42709e-9	-8.51969e-18	-4.42709e-9	4.42641e-9	7.76564e-11	1	1.10677e-6
6.46086	1.52743	7.37316	-0.0175412	-0.999846	8.68658	500	-4.40595e-6	-500	499.923	8.77058	4.42709e-9	-3.90111e-17	-4.42709e-9	4.42641e-9	7.76563e-11	1	1.10677e-6

Comprobamos $\mathbf{Q_s}$ con Python.

```python
# Constantes
eps0 = 8.8541878128e-12  # F/m

# Tus datos (en metros)
ancho_total = 4e-3       # 4 mm
alto = 20e-3             # 20 mm
Lz = 10e-3               # 10 mm de profundidad
V = 10                   # Voltios

# Cada mitad tiene la mitad del ancho
ancho_mitad = ancho_total / 2
area_mitad = ancho_mitad * Lz  # m²
d = alto  # separación = alto del rectángulo

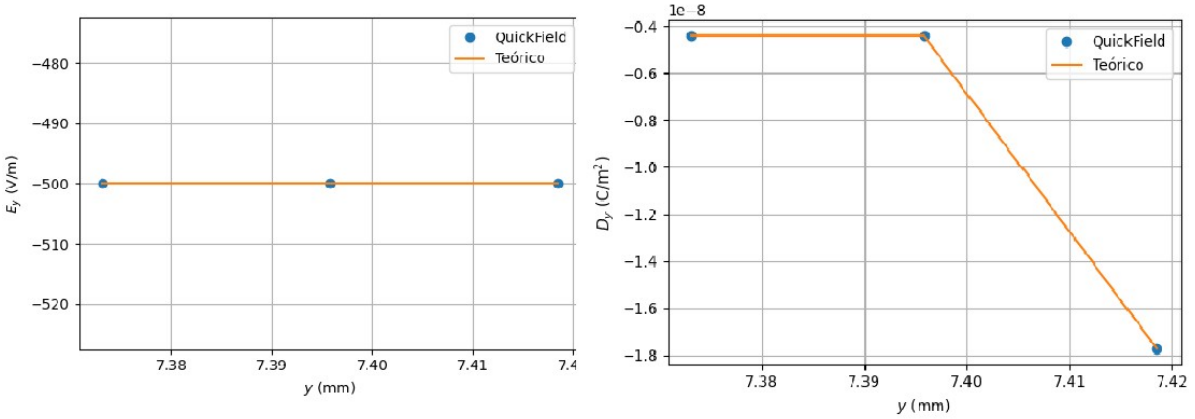
# Materiales
eps_r1 = 4  # izquierda
eps_r2 = 1  # derecha

# Cálculo
C1 = eps0 * eps_r1 * (area_mitad / d)
C2 = eps0 * eps_r2 * (area_mitad / d)
C_total = C1 + C2
Q_teorico = C_total * V

# Resultados
print(f"C1 = {C1:.3e} F")
print(f"C2 = {C2:.3e} F")
print(f"C_total = {C_total:.3e} F")
print(f"Q_teórico = {Q_teorico:.3e} C")
print(f"Q_simulado = 4.427e-13 C")
print(f"Diferencia = {abs(Q_teorico - 4.427e-13):.2e} C")
```

```
C1 = 3.542e-14 F
C2 = 8.854e-15 F
C_total = 4.427e-14 F
Q_teórico = 4.427e-13 C
Q_simulado = 4.427e-13 C
Diferencia = 9.39e-18 C
```

Gráficas del **desplazamiento** y el **campo eléctrico.**



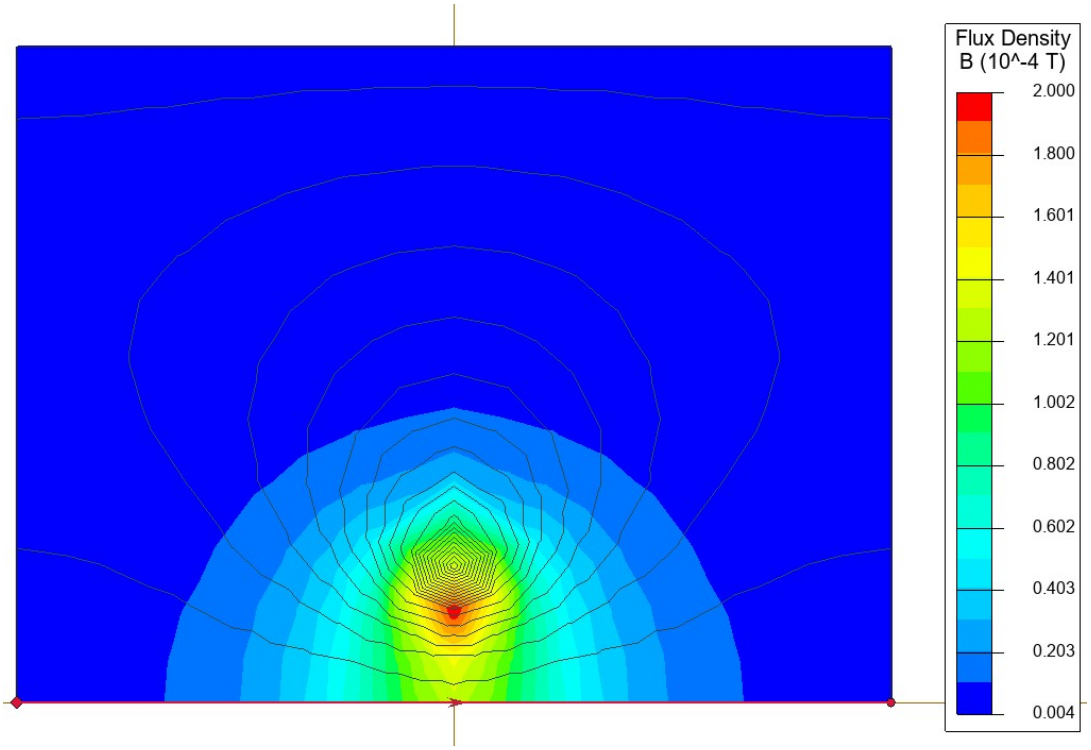
3.1. Conclusión.

Los resultados obtenidos mediante QuickField muestran un valor prácticamente constante de D_y, en concordancia con la teoría, lo que confirma la correcta modelización del sistema.

4. Espira Corriente.

Configuración en **QuickField**:

- $w = 80$
- $h = 60$
- $y = 30$
- Radio $R = 12.5$
- Corriente $I = 2.5$



Obtenemos la tabla de datos.

L, mm	z, mm	r, mm	N_x	N_y	rA, Wb	B, T	B_z, T	B_r, T	B_n, T	B_t, T	H, A/m	H_z, A/m	H_r, A/m	H_n, A/m	H_t, A/m	μ	w, J/m^3
0	-40	0	0	-1	0	5.40782e-6	5.40782e-6	0	0	5.40782e-6	4.30341	4.30341	0	0	4.30341	1	1.1636e-5
4	-36	0	0	-1	0	5.78843e-6	5.78831e-6	-3.60943e-8	3.60943e-8	5.78831e-6	4.60628	4.60619	-0.028723	0.028723	4.60619	1	1.33316e-5
8	-32	0	0	-1	0	6.83636e-6	6.83543e-6	-1.12739e-7	1.12739e-7	6.83543e-6	5.4402	5.43946	-0.0897147	0.0897147	5.43946	1	1.85956e-5
12	-28	0	0	-1	0	8.92489e-6	8.92283e-6	-1.91547e-7	1.91547e-7	8.92283e-6	7.1022	7.10056	-0.152428	0.152428	7.10056	1	3.16932e-5
16	-24	0	0	-1	0	1.27752e-5	1.27671e-5	-4.52966e-7	4.52966e-7	1.27671e-5	10.1662	10.1598	-0.360459	0.360459	10.1598	1	6.49372e-5
20	-20	0	0	-1	0	1.94358e-5	1.94273e-5	-5.74147e-7	5.74147e-7	1.94273e-5	15.4665	15.4598	-0.456892	0.456892	15.4598	1	1.50302e-4
24	-16	0	0	-1	0	3.02556e-5	3.02379e-5	-1.03253e-6	1.03253e-6	3.02379e-5	24.0766	24.0626	-0.821657	0.821657	24.0626	1	3.64226e-4
28	-12	0	0	-1	0	4.81649e-5	4.81341e-5	-1.72264e-6	1.72264e-6	4.81341e-5	38.3284	38.3039	-1.37083	1.37083	38.3039	1	9.23041e-4
32	-8	0	0	-1	0	7.53555e-5	7.5312e-5	-2.558e-6	2.558e-6	7.5312e-5	59.966	59.9314	-2.03559	2.03559	59.9314	1	0.00225938
36	-4	0	0	-1	0	1.05694e-4	1.05674e-4	-2.03286e-6	2.03286e-6	1.05674e-4	84.1085	84.093	-1.6177	1.6177	84.093	1	0.00444488
40	1.73472e-15	0	0	-1	0	1.26388e-4	1.26388e-4	0	0	1.26388e-4	100.576	100.576	0	0	100.576	1	0.00635582
44	4	0	0	-1	0	1.05694e-4	1.05674e-4	2.03286e-6	-2.03286e-6	1.05674e-4	84.1085	84.093	1.6177	-1.6177	84.093	1	0.00444488
48	8	0	0	-1	0	7.53555e-5	7.5312e-5	2.558e-6	-2.558e-6	7.5312e-5	59.966	59.9314	2.03559	-2.03559	59.9314	1	0.00225938
52	12	0	0	-1	0	4.81649e-5	4.81341e-5	1.72264e-6	-1.72264e-6	4.81341e-5	38.3284	38.3039	1.37083	-1.37083	38.3039	1	9.23041e-4
56	16	0	0	-1	0	3.02556e-5	3.02379e-5	1.03253e-6	-1.03253e-6	3.02379e-5	24.0766	24.0626	0.821658	-0.821658	24.0626	1	3.64226e-4
60	20	0	0	-1	0	1.94358e-5	1.94273e-5	5.74147e-7	-5.74147e-7	1.94273e-5	15.4665	15.4598	0.456892	-0.456892	15.4598	1	1.50302e-4
64	24	0	0	-1	0	1.27752e-5	1.27671e-5	4.52966e-7	-4.52966e-7	1.27671e-5	10.1662	10.1598	0.360459	-0.360459	10.1598	1	6.49372e-5
68	28	0	0	-1	0	8.92489e-6	8.92283e-6	1.91547e-7	-1.91547e-7	8.92283e-6	7.1022	7.10056	0.152428	-0.152428	7.10056	1	3.16932e-5
72	32	0	0	-1	0	6.83636e-6	6.83543e-6	1.12739e-7	-1.12739e-7	6.83543e-6	5.4402	5.43946	0.0897147	-0.0897147	5.43946	1	1.85956e-5
76	36	0	0	-1	0	5.78842e-6	5.78831e-6	3.60943e-8	-3.60943e-8	5.78831e-6	4.60628	4.60619	0.0287229	-0.0287229	4.60619	1	1.33316e-5
80	40	0	0	-1	0	5.40782e-6	5.40782e-6	0	0	5.40782e-6	4.30341	4.30341	0	0	4.30341	1	1.1636e-5

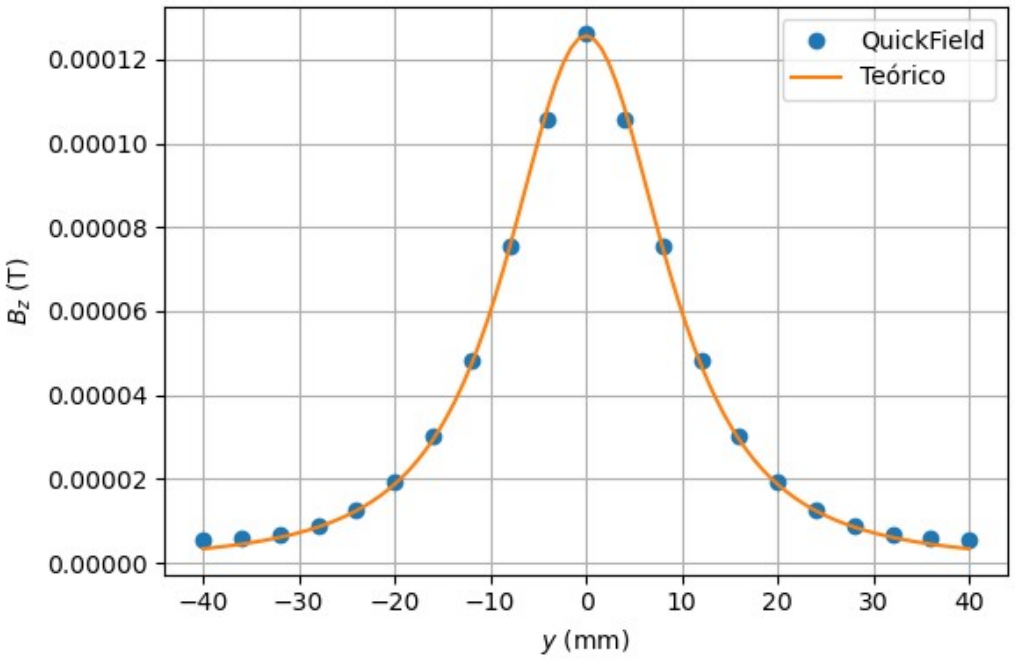
Modificamos los valores en Python.

```
# (para problema magnetostático)
data = pd.read_csv(file_name, sep=r"\s+", skiprows=1,
                   names=["L_mm", "z_mm", "r_mm", "Nx", "Ny", "rA_Wb", "B_T",
                          "Bz_T", "Br_T", "Bn_T", "Bt_T", "H_A/m",
                          "Hz_A/m", "Hr_A/m", "Hn_A/m", "Ht_A/m", "mu_r", "w_J/m3"])

################################################################################
# Extraer columnas de interés
L = data["z_mm"]
Bz = data["Bz_T"]

# Modelo teórico: campo de una espira circular de corriente
mu0 = 4*np.pi*1e-7  # permeabilidad del vacío
I_espira = 2.5  # A
z = np.linspace(np.min(L), np.max(L), 101)
R = 12.50  # radio de la espira [mm]
Bz_teo = mu0*I_espira*(R*1e-3)**2 / 2 / ((z*1e-3)**2 + (R*1e-3)**2)**(3/2)
```

Obtenemos el Plot del código.



4.1. Conclusión.

Los resultados obtenidos mediante QuickField muestran una concordancia con la solución teórica. Se comprobo con Bz_{teo} en Python, que la componente axial del campo magnético sigue la depencia teórica.

5. Revisión.


Tu calificación
Calificado por
JORGE GINER NAVARRO
Fecha y hora de calificación
18/12/2025 15:31
Tu nota actual
10,00
/ 10,00